Sara Latreche

Contents

1 The Genesis of Data Analysis: A Historical Shift **5**

1.1 From Probability to Applied Science 5

1.2 Classical Inference vs. Modern EDA 5

1.3 The Tukey Revolution: Visuals and Language 7

1.4 Why EDA Matters for Data Scientists 8

1.5 The Taxonomy of Statistics 8

2 Elements of Structured Data **9**

2.1 Key Data Types 9

2.2 Rectangular Data: The Data Frame 9

2.3 Python vs. R: Indexing 10

2.4 Visual Overview: Elements of Structured Data 11

3 Estimates of Location: Finding the Center **11**

3.1 The Mean (Average) 11

3.2 Trimmed Mean 12

3.3 Weighted Mean 12

3.4 The Median (50th Percentile) 12

3.5 Weighted Median 12

3.6 Outliers and Robustness 12

4 Estimates of Variability: Measuring the Spread **13**

4.1 The Classical Metrics (Sensitive to Outliers) 13

4.2 The Robust Metrics (Tukey's Favorites) 14

5 Visualizing the Distribution: Beyond the Numbers **14**

5.1 Key Tools for Visual Discovery 14

5.2 The Four Statistical Moments 15

6 Intuitive Data Exploration: From Histograms to Density **15**

6.1 The Path to Probability Density (KDE) 16

6.2 The Anatomy of a Box Plot 17

6.3 The Statistical Definition of Outliers 18

7 The Evolution of Visualization: The Violin Plot **18**

7.1 Building a Violin Plot from Scratch 18

7.2 The Mathematics of Kernel Density Estimation (KDE) 18

8 Comparing Visual Distillations **19**

8.1 1. The Histogram: The Raw Density 20

8.2 2. The Box Plot: The Five-Number Summary 20

8.3 3. The Violin Plot: The Best of Both Worlds 21

8.4 Which Plot Should You Use? 22

8.5 Practical Comparison: Which Spread to Trust? 23

9 Exploring Binary and Categorical Data **23**

9.1 Proportions and Percents 23

9.2 The Expected Value 24

10 Covariance and Correlation **24**

10.1 Covariance: The Direction of Change 24

10.1.1 Covariance Formulas 25

10.2 Correlation: The Standardized Solution 25

10.2.1 General Correlation Formulas 25

10.3 Types of Correlation Coefficients 26

10.4 Pearson's Correlation Coefficient (r) 26

10.4.1 Manual Calculation Formula 27
10.4.2 Numerical Example: Age vs. Income 27
10.5 Spearman's Rank Correlation Coefficient (ρ) 28
10.5.1 Spearman Formula 28
10.5.2 Numerical Example: History vs. Geography 29

11 Advanced Correlation Analysis **31**
11.1 Verification via Scatter Plots 31
11.2 The Correlation Heatmap 33
11.3 Choosing the Best Method: Pearson vs. Spearman 34
11.4 Correlation vs. Causation 34

12 Exploring Two or More Variables **34**
12.1 Visualizing Numeric vs. Numeric Data 35
12.1.1 Hexagonal Binning 35
12.1.2 Contour Plots (Topographical Maps) 35

13 Multicollinearity and VIF **36**
13.1 Variance Inflation Factor (VIF) 36
13.2 Interpretation Guide 37

14 Data and Sampling Distributions **38**
14.1 Population versus Sample 38

15 Random Sampling and Sample Bias **39**
15.1 Key Terms 39
15.2 The Importance of Data Quality 39

16 Selection Bias and Regression to the Mean **39**
16.1 Selection Bias 39
16.1.1 Bias vs. Random Error 39

16.2 Regression to the Mean 40

17 Sampling Distribution of a Statistic **40**

17.1 The Central Limit Theorem (CLT) 40

17.2 Visualizing Convergence 41

17.3 The Mathematical Formula 41

18 The Bootstrap **42**

18.1 Concept and Procedure 43

18.2 Bootstrap Aggregating (Bagging) 43

19 Confidence Intervals **43**

19.1 Visualizing the Interval 44

19.2 Key Drivers of Interval Width 44

19.3 Key Ideas 45

20 Important Distributions in Data Science **45**

20.1 Key Distributions and Their Utility 45

20.2 Important Distributions in Data Science 46

20.3 Key Ideas 46

20.4 Quantifying Difference: KL Divergence 47

21 Inferential Statistics: The Signal and the Noise **48**

21.1 Why We Use Statistical Tests 48

21.2 Choosing Your "Signal Detector" 48

22 Bayesian Statistics **49**

22.1 Bayes' Theorem 50

22.2 Utility in Data Science 50

1 The Genesis of Data Analysis: A Historical Shift

Before diving into Python code, we must understand the "why" behind our tools. Statistics is a relatively young discipline, and its evolution mirrors the evolution of Data Science itself.

1.1 From Probability to Applied Science

While the mathematical foundations of probability were laid by giants like **Bayes**, **Laplace**, and **Gauss** between the 17th and 19th centuries, it remained largely theoretical. It wasn't until the late 1800s, with **Francis Galton** and **Karl Pearson**, that statistics emerged as a rigorous applied science.

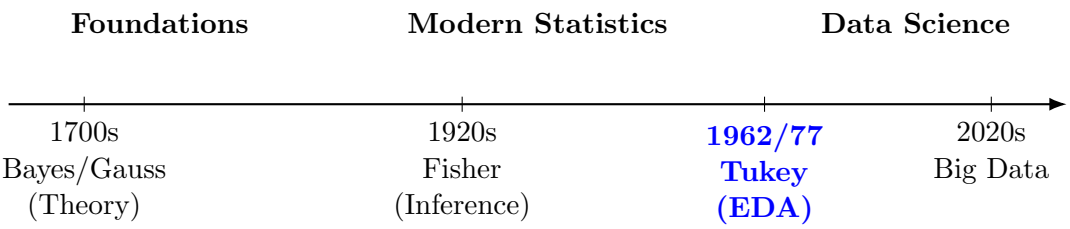

Figure 1: The evolution from theoretical probability to exploratory data analysis.

The Pioneers of Modern Statistics

- **R.A. Fisher (Early 20th Century):** Pioneered experimental design and *Maximum Likelihood Estimation.*
- **John W. Tukey (1960s-70s):** Often called the "Father of Data Science," he revolutionized how we look at information by moving away from traditional, rigid mathematical statistics.

1.2 Classical Inference vs. Modern EDA

Traditionally, statistics was obsessed with **Inference**: the art of drawing conclusions about a large population based on a very small sample. This was a necessity in an era where data was expensive and hard to collect.

Before Tukey, statistics was mostly **Confirmatory Data Analysis (CDA)**. This meant you had to have a specific hypothesis first and then use complex math to prove if it was right or wrong (using p-values and significance tests).

Classical Inference (CDA)

- Small, Expensive Samples
- Formal Hypothesis Testing
- P-values & Significance

$\longrightarrow$

Modern EDA

- Large, Complex Datasets
- Pattern & Clue Discovery
- Visual Intuition & Iteration

In 1962, Tukey published *"The Future of Data Analysis"*, calling for a reformation. He argued that data analysis should be treated as an **empirical science** rather than just a branch of mathematics. He introduced **Exploratory Data Analysis (EDA)**—the act of looking at data to see what it can tell us beyond the formal modeling task.

Tukey believed:

- You should look at the data first to see what it is telling you before making assumptions.
- Data analysis is like **detective work**—you are looking for clues and patterns rather than just trying to prove a point.
- An "approximate answer to the right question" is much better than an "exact answer to the wrong question."

Note

As David Donoho (Stanford) noted, the genesis of modern Data Science is not found in a new algorithm, but in Tukey's shift toward data-driven discovery rather than just hypothesis-driven proof.

1.3 The Tukey Revolution: Visuals and Language

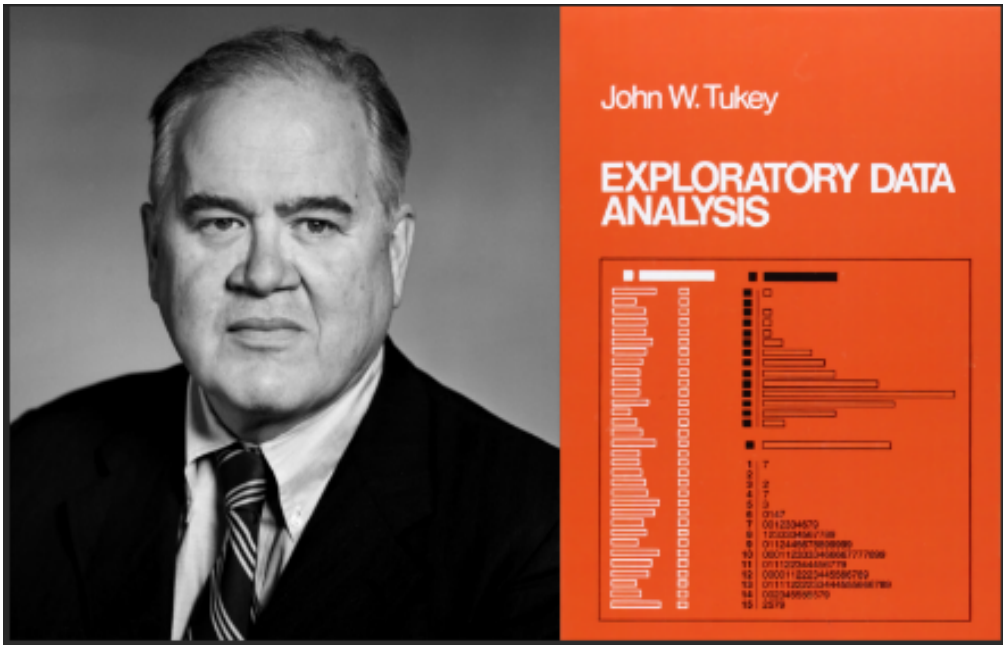

Figure 2: John W. Tukey and the cover of his landmark 1977 book, *Exploratory Data Analysis*.

Tukey believed that the human eye is one of the best tools for finding patterns. He invented simple, visual ways to "see" data that didn't require a PhD in math to understand:

- **The Box Plot (Box-and-Whisker):** A simple way to see the spread of data and find outliers.
- **The Stem-and-Leaf Display:** A way to see the actual numbers while also seeing the shape of the distribution.
- **Graph Paper over Equations:** He famously said that EDA is an "attitude" and "some graph paper," emphasizing flexibility over rigid formulas.

He also pushed for **"Robust" statistics**—methods that still work even if the data is a bit messy or doesn't follow a perfect bell curve. He felt that traditional statisticians were too focused on "mathematical purity" and not enough on the "test of experience."

Finally, because he saw the future of data being tied to computers, he gave us the vocabulary we use today. He coined the term **"Bit"** (Binary Digit) and is credited with the first published use of the word **"Software."**

1.4 Why EDA Matters for Data Scientists

In the age of "Big Data," the risk is no longer having "too little" data, but being overwhelmed by its complexity. EDA allows us to:

1. **Verify Assumptions:** Is the data actually distributed the way our model expects?
2. **Identify Anomalies:** Are there outliers that will skew our results?
3. **Feature Engineering:** Which variables actually hold predictive power?

Active Recall – History Check

John Tukey coined a term we use every single day in computing. Based on the text, what are the two terms he is credited with creating?

1.5 The Taxonomy of Statistics

Modern data science utilizes various branches of statistics depending on the business question being asked. We can categorize these into four primary types:

Type	Goal	Common Tools
Descriptive	Summarize what happened in the past.	Mean, SD, Histograms
Inferential	Generalize from a sample to a population.	T-tests, P-values
Predictive	Forecast what might happen next.	Regression, Random Forest
Prescriptive	Recommend a specific course of action.	Optimization, Simulation

Table 1: The four pillars of statistical analysis in Data Science.

Example

In the **California Housing** dataset:

- **Descriptive:** What is the average house price in California?
- **Inferential:** Does the sample of 20,000 homes represent the entire state accurately?
- **Predictive:** Can we predict the price of a new house based on its location and age?

Note

The Data Scientist's Edge: ML and Deep Learning
While classical statistics often focuses on linear relationships and small datasets, Data Scientists extend these concepts into:

- **Machine Learning (ML):** Using statistical algorithms to "learn" patterns from data without being explicitly programmed. It is essentially *Predictive Statistics* at scale.
- **Deep Learning (DL):** A subset of ML inspired by the human brain (Neural Networks) that excels at unstructured data like images or text.

In this course, we treat Statistics as the **engine** and Machine Learning as the **vehicle** that takes us to our destination.

2 Elements of Structured Data

In the era of Big Data, we often deal with raw, "unstructured" information (text, images, video). However, for statistical modeling, this must be transformed into **Structured Data**.

2.1 Key Data Types

Statistical algorithms require us to categorize data based on its nature. This dictates which charts we use and which models are appropriate.

Data Type	Description	Synonyms
Continuous	Any numeric value in a range.	float, numeric, interval
Discrete	Integer values (counts).	count, integer
Categorical	Limited categories/labels.	enums, nominal, factors
Binary	Only two categories (0/1).	dichotomous, indicator
Ordinal	Categories with a natural order.	ordered factor

Table 2: Taxonomy of Data Types in Data Science.

2.2 Rectangular Data: The Data Frame

The fundamental unit of data science is the **Data Frame**: a rectangular grid where rows represent observations and columns represent variables.

Category	Currency	SellerRating	ClosePrice
Music	US	3249	0.01
Automotive	US	3115	0.01

← **Features** (Columns/Predictors)

← **Outcome** (Target Variable)

Record (Row/Instance)

Terminology Bridge

Different fields use different names for the same things. In this course, we will use these interchangeably:

- **Feature** = Attribute = Predictor = Input.
- **Record** = Case = Instance = Sample.
- **Outcome** = Target = Dependent Variable = Response.

2.3 Python vs. R: Indexing

Note

Python (pandas) uses zero-based indexing by default. The `DataFrame` object is the standard. **R** uses one-based indexing. While `data.frame` is built-in, libraries like `data.table` are used for high-performance big data tasks.

Active Recall – Concept Mapping

In a dataset tracking breast cancer patients, would "Tumor Size" be a Continuous or Discrete variable? What about "Malignant/Benign"?

2.4 Visual Overview: Elements of Structured Data

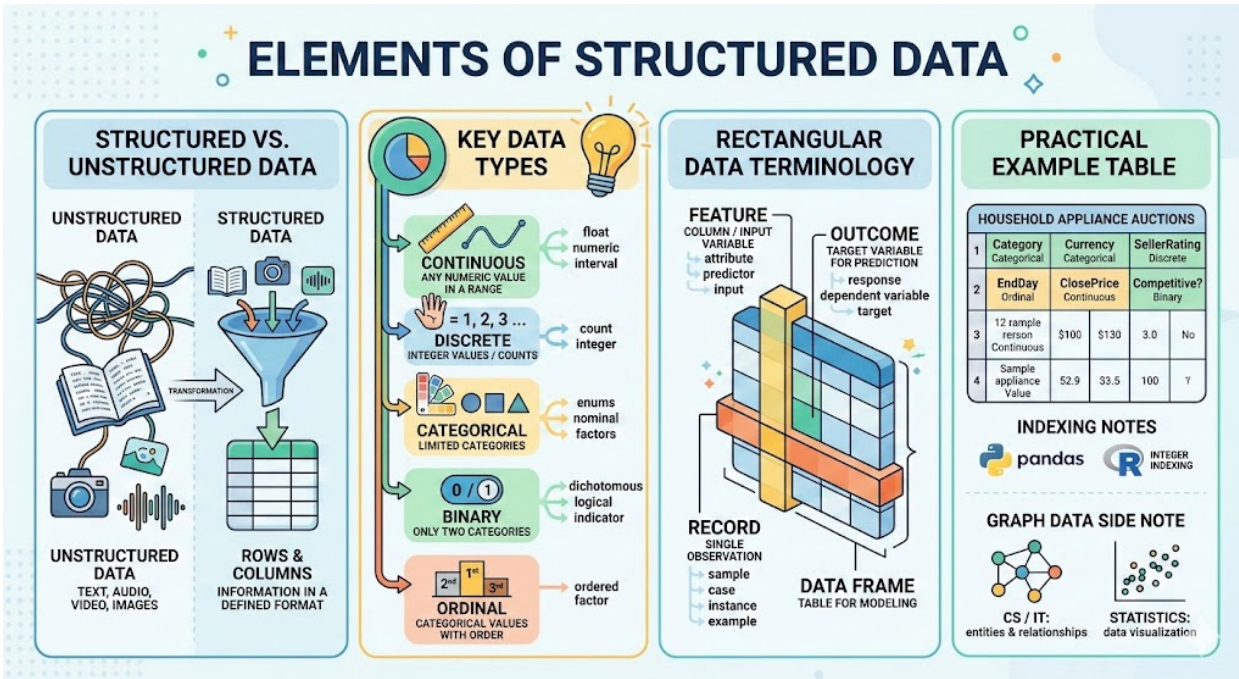

Figure 3: Comprehensive breakdown of data types and rectangular structures.

Note

The infographic above summarizes how unstructured data (like images/text) is funneled into the structured rows and columns we use for statistical analysis.

3 Estimates of Location: Finding the Center

When analyzing numerical data, we often want to know where the values are "centered." These are called **estimates of location**. In the spirit of Tukey, we must choose our "center" based on how messy our data is.

3.1 The Mean (Average)

The mean is the most common estimate, calculated as the sum of all values divided by the number of values:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

3.2 Trimmed Mean

To reduce the impact of outliers, we can calculate the mean after removing the p smallest and p largest values from the sorted dataset:

$$\text{Trimmed Mean} = \frac{1}{N-2p} \sum_{i=p+1}^{N-p} x_i$$

3.3 Weighted Mean

Useful when some values (x_i) are more important or reliable than others, each is assigned a weight (w_i):

$$\bar{x}_w = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i}$$

3.4 The Median (50th Percentile)

The median is the middle value in a **sorted** dataset.

- If N is odd: The middle value.
- If N is even: The average of the two middle values.

The median is a **robust estimate**; unlike the mean, it is not "pulled" by extreme values.

[Image of mean vs median in skewed distribution]

3.5 Weighted Median

Similar to the median, the weighted median is the point in the sorted data where the cumulative weight reaches 50%. It remains robust while accounting for the importance of specific data points.

3.6 Outliers and Robustness

Understanding the difference between metrics is critical for a Data Scientist:

- **Outlier:** A value far from the rest of the data (often caused by errors or rare events).
- **Robust Estimate:** A metric (like the **median** or **trimmed mean**) that is not sensitive to these extreme values.

Tool	Best Use Case	Robustness
Mean	Clean, symmetric data	Low (Sensitive)
Trimmed Mean	Data with few extreme outliers	Medium
Median	Skewed data or heavy outliers	High (Robust)

Table 3: Choosing the right estimate of location.

Key Insight

Use the **mean** when the data is clean and follows a normal distribution. Use the **median** or **trimmed mean** when the data is "dirty," skewed, or contains anomalies.

4 Estimates of Variability: Measuring the Spread

Location is only one side of the story; **variability** (or dispersion) is the second. It measures whether your data points are tightly clustered or widely spread out. In Data Science, understanding spread is the key to identifying uncertainty and anomalies.

4.1 The Classical Metrics (Sensitive to Outliers)

These are the standard measures taught in introductory statistics. While powerful, they are highly sensitive to extreme values because they rely on the **mean**.

Variance (s^2): Measures the average squared distance from the mean.

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

Standard Deviation (s): The square root of variance. It is the most popular metric because it is expressed in the same units as the original data.

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

Mean Absolute Deviation (Mean AD): The average of the absolute distances from the mean.

$$\text{Mean AD} = \frac{1}{N}\sum_{i=1}^{N}|x_i - \bar{x}|$$

4.2 The Robust Metrics (Tukey's Favorites)

Just as the median is a robust estimate of location, these tools provide a stable measure of spread that isn't ruined by a few "wild" data points.

Median Absolute Deviation (MAD): Tukey's preferred "robust" version of standard deviation. It calculates the median distance of all points from the median (m).

$$\text{MAD} = \text{Median}(|x_1 - m|, |x_2 - m|, \ldots, |x_N - m|)$$

Interquartile Range (IQR): The IQR represents the spread of the **middle 50%** of the data. To calculate it, we subtract the 25th percentile (Q1) from the 75th percentile (Q3):

$$\text{IQR} = P_{75} - P_{25}$$

Note

The 1.5 × IQR Rule: Tukey used the IQR to define outliers mathematically. Any data point that falls more than $1.5 \times \text{IQR}$ above the 75th percentile or below the 25th percentile is flagged as a potential outlier.

Active Recall – Why use MAD or IQR?

Standard Deviation is like a rubber band: one outlier can stretch it to the breaking point. **MAD** and **IQR** are like a steel ruler: they stay the same size regardless of what happens at the extreme edges of your dataset.

5 Visualizing the Distribution: Beyond the Numbers

While location and variability summarize data with single values, they can sometimes hide the true nature of the data. To truly understand our dataset, we must explore its **distribution**—the pattern of how often different values occur.

5.1 Key Tools for Visual Discovery

In the spirit of Tukey's "Data Detective" philosophy, we use these four primary tools to "see" our data:

- **Boxplot (Box and Whiskers):** A graphical summary introduced by Tukey to visualize the 5-number summary (Min, Q1, Median, Q3, Max) and identify outliers at a glance.

- **Frequency Table:** A tabular count of how many data points fall into specific numeric intervals (bins).
- **Histogram:** A bar chart of the frequency table. It shows the "shape" of the data by plotting bins on the x-axis and counts on the y-axis.
- **Density Plot:** A smoothed version of the histogram, often created using *Kernel Density Estimation* (KDE). It helps visualize the distribution's shape without being distracted by bin sizes.

5.2 The Four Statistical Moments

In statistics, we describe the "personality" of a distribution using four mathematical **moments**. Each one tells us something specific about the shape:

1. **1st Moment – Mean (Location):** Where the "balance point" of the data is located.
2. **2nd Moment – Variance (Variability):** How far the data spreads out from that center.
3. **3rd Moment – Skewness:** Measures the asymmetry.
   - *Positive Skew:* A long tail to the right.
   - *Negative Skew:* A long tail to the left.
4. **4th Moment – Kurtosis:** Measures "peakedness" and the thickness of the tails. High kurtosis means the data has frequent, extreme outliers (heavy tails).

Active Recall – Tukey's Reminder

A single number like the **mean** can be the same for two completely different datasets. Always plot your data (using a Histogram or Boxplot) to see the "moments" in action before you start modeling.

6 Intuitive Data Exploration: From Histograms to Density

Building on Tukey's "Data Detective" mindset, we move from raw numbers to visual patterns. Consider a simple sample dataset:

$$[2.1, 2.4, 2.5, 2.6, 3.0, 3.1, 3.1, 3.3, 3.8, 4.0]$$

6.1 The Path to Probability Density (KDE)

To see where these values are most concentrated, we transition from rigid blocks to smooth curves.

1. The Histogram (The Rough Shape): We start by dividing the number line into "bins" and counting how many values fall into each. It is essentially like stacking blocks where more data appears.

2. Kernel Density Estimation (The Smooth Curve): KDE takes the histogram further by placing a smooth "bump" (a mini bell curve) over each individual data point. When these hills are added together, they form a smooth distribution curve that shows where the data "lives."

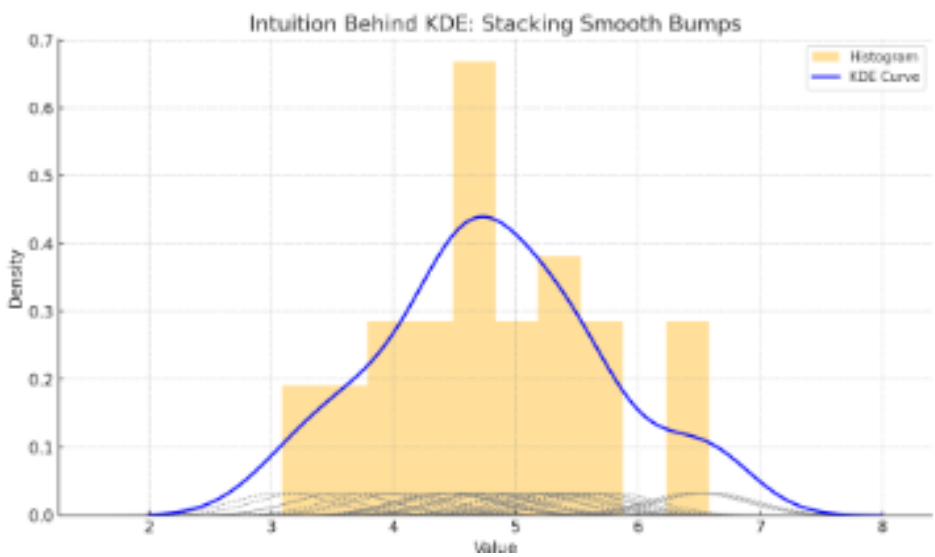

Figure 4: Intuition Behind KDE: Stacking smooth bumps over data points to create a continuous density curve.

Why is it called "Density"?

- It shows the **relative likelihood** of values.
- The total **area under the curve** is always 1 (100% probability).
- It represents the shape the data would form if we had infinitely many points.

6.2 The Anatomy of a Box Plot

A box plot is a graphical summary of a dataset's "Five-Number Summary." It is the most efficient way to visualize the spread and skewness of a feature at a glance.

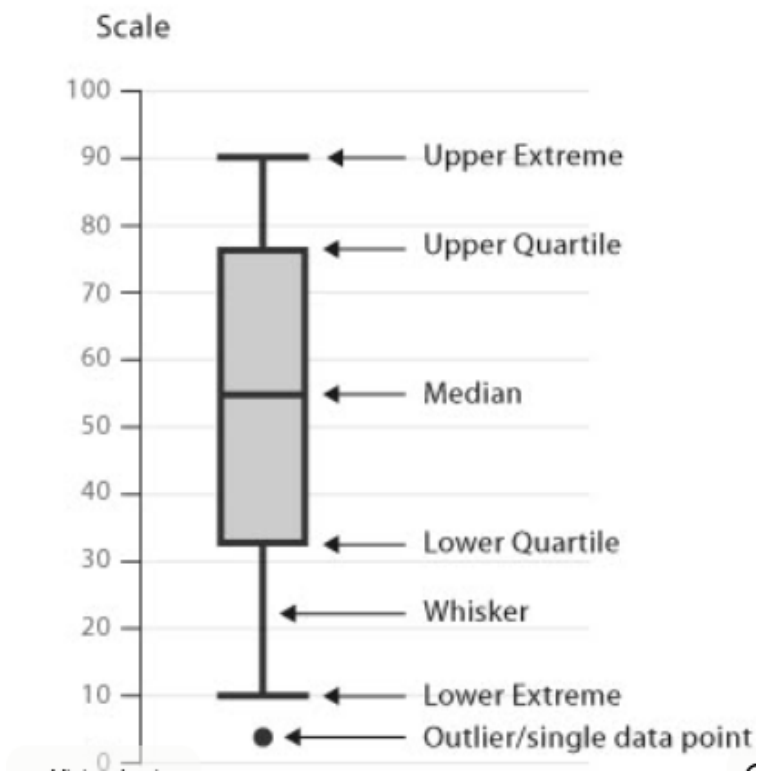

Figure 5: Box Plot Explanation: Visualizing the quartiles, median, and detected outliers.

Key Components:

- **The Box:** Represents the Interquartile Range (IQR), containing the middle 50% of the data.
- **The Median (Q2):** The red dashed line marking the 50th percentile.
- **The Whiskers:** Lines extending to the minimum and maximum values (excluding outliers).
- **Outliers:** The individual circles plotted far to the right, representing values that fall outside the typical range.

6.3 The Statistical Definition of Outliers

Tukey provided a rigorous mathematical "fence" to identify these anomalies. An **outlier** is any data point that lies outside the following range:

$$[Q_1 - 1.5 \times IQR, \quad Q_3 + 1.5 \times IQR]$$

Active Recall – Concept Check

Looking at the Box Plot above, the values at 30 and 35 are flagged as outliers. Even though they pull the **mean** higher, the **median** (12) stays firmly in the center of the data. This is why Tukey championed these robust measures!

7 The Evolution of Visualization: The Violin Plot

If a box plot summarizes the data and a KDE plot shows its shape, a **Violin Plot** provides the best of both worlds. It is the ultimate tool for comparing distributions across different categories.

7.1 Building a Violin Plot from Scratch

An intuitive way to understand a violin plot is to see it as a hybrid construction:

1. **Raw Data:** Start with the individual observations.
2. **KDE Estimation:** Estimate where the data is concentrated using Kernel Density Estimation.
3. **Mirroring:** Mirror that KDE curve around a vertical axis to create the "violin" body.
4. **Integration:** Place a miniature box plot inside the center to show quartiles and the median.

The Violin Equation

$$\text{Violin Plot} = \text{Box Plot} + \text{Mirrored KDE}$$

7.2 The Mathematics of Kernel Density Estimation (KDE)

To understand how the "violin" gets its shape, we look at the underlying math. The KDE estimates the **probability density function (PDF)** of the dataset by placing a "bump" (kernel) at each data point and summing them.

The formula for the estimated density $\hat{f}_h(x)$ is:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

Where:

- n: Total number of observations.
- h: The **Bandwidth** parameter (controls how smooth or "wiggly" the curve is).
- x_i: The ith data point in the sample.
- $K(\cdot)$: The **Kernel function**.

For a **Gaussian kernel**, which creates the familiar smooth "bell" shape for each point, we use:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}$$

Note

While the box plot is excellent for spotting outliers, the violin plot excels at showing **multimodality** (when a dataset has two or more "peaks"). A box plot might hide the fact that your data has two distinct clusters, but the violin plot will clearly show two "fat" sections.

Active Recall – Final Summary Checklist

- Use a **Box Plot** when you need to identify outliers and quartiles quickly.
- Use a **KDE Plot** when you want to see the smooth probability density of a single feature.
- Use a **Violin Plot** when you want to compare the full distribution shape between multiple groups.

8 Comparing Visual Distillations

In data science, we often have to choose between simplicity and detail. By looking at the **Texture Error** from the Breast Cancer dataset, we can see how Histograms, Box plots, and Violin plots each tell a different story.

8.1 1. The Histogram: The Raw Density

The histogram is our most basic tool for seeing frequency. It shows us exactly where the "piles" of data are.

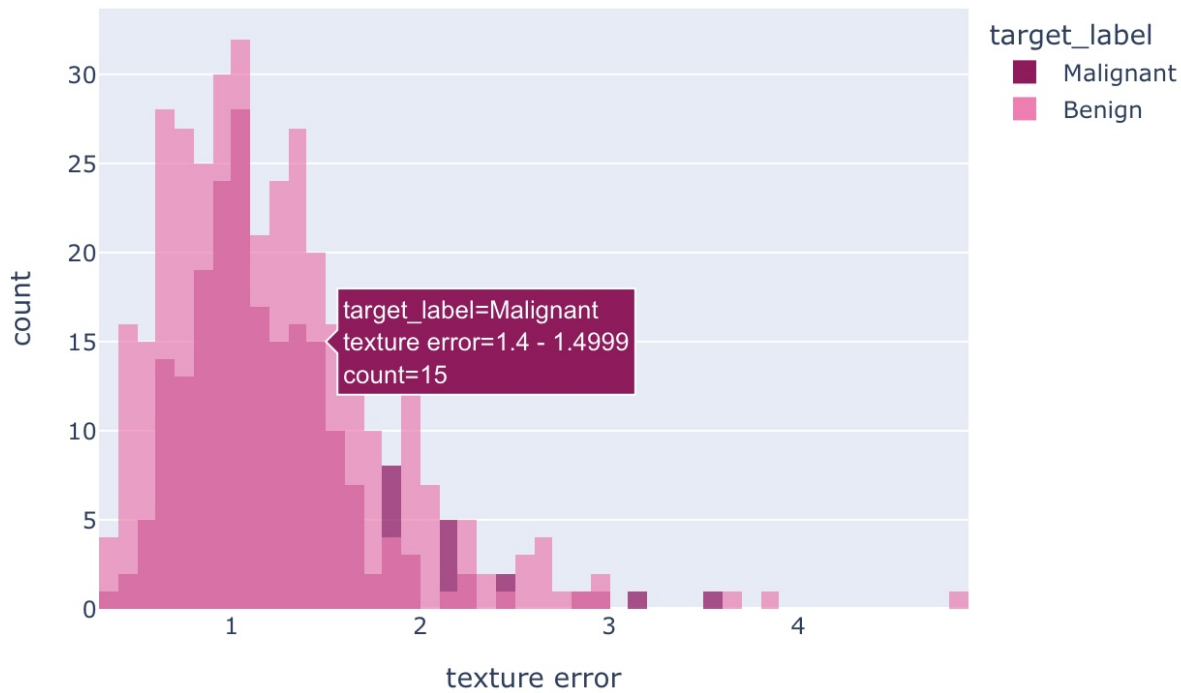

Figure 6: Histogram of Texture Error: Malignant vs. Benign.

Note

Notice the overlap. The histogram reveals that while both diagnoses peak around a texture error of 1.0, the **Benign** cases (light pink) have a much broader base and several small "islands" of data points further to the right.

8.2 2. The Box Plot: The Five-Number Summary

While the histogram shows every "bump," the Box Plot distills the data into five critical summary statistics.

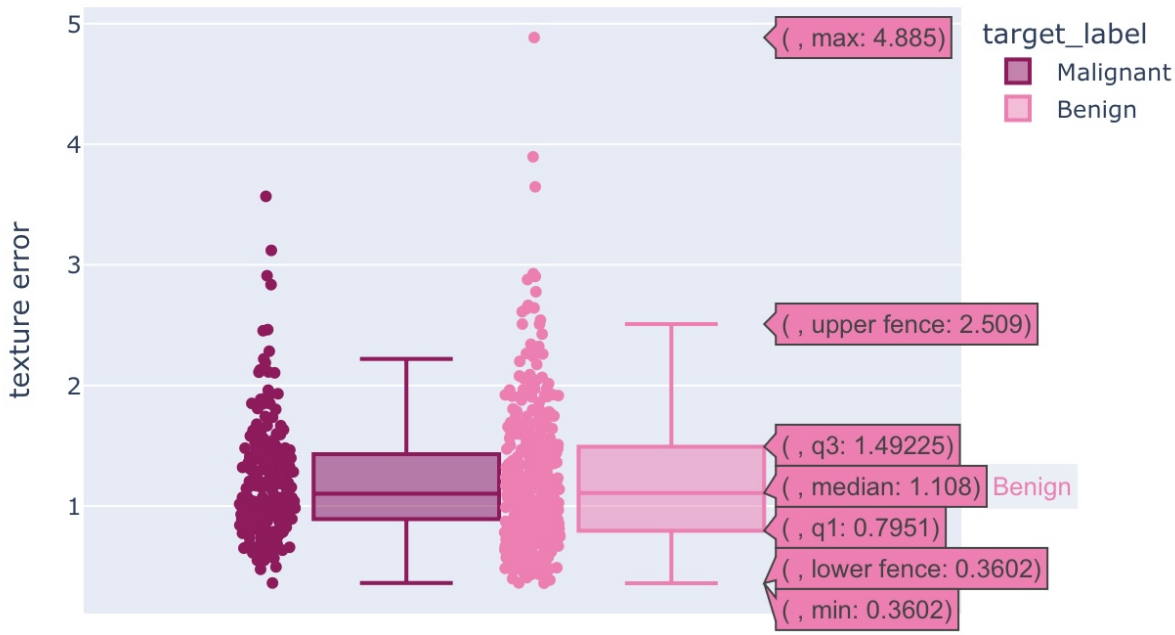

Figure 7: Annotated Box Plot: Visualizing Quartiles and Fences.

Reading the Box Plot

As seen in the annotations:

- **Median (Q2):** The center line (1.108 for Benign).
- **IQR (Q1 to Q3):** The "box" containing the middle 50% of the data.
- **Outliers:** The individual dots beyond the "upper fence" (2.509).

8.3 3. The Violin Plot: The Best of Both Worlds

The violin plot is the modern standard for data scientists. It provides the statistical rigor of a box plot with the visual density of a histogram.

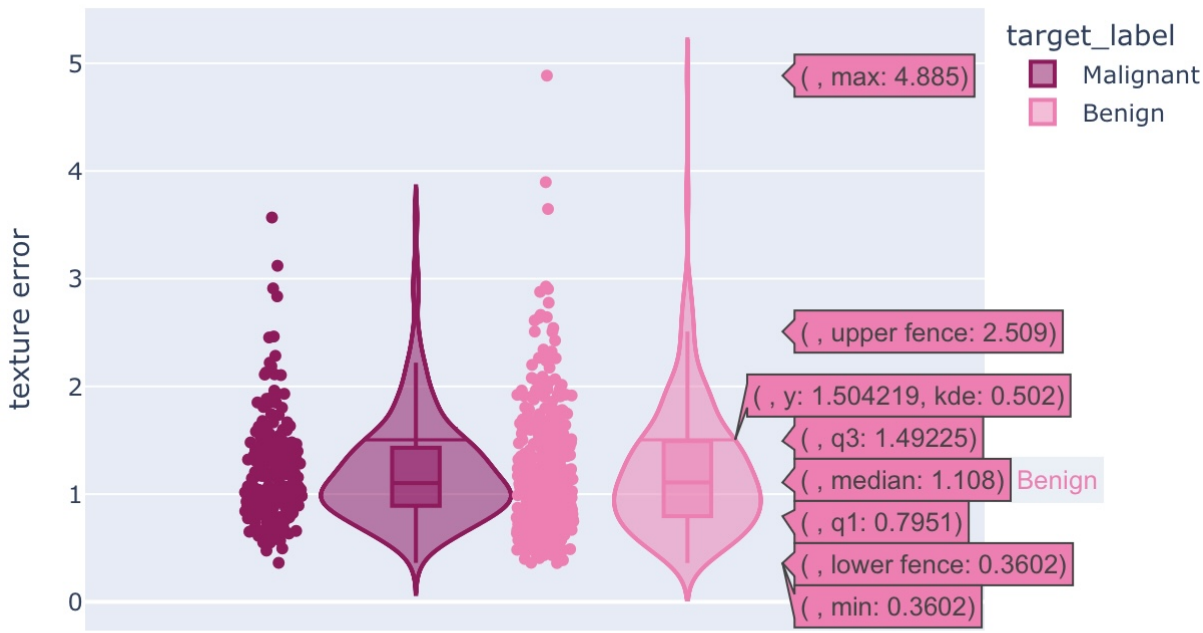

Figure 8: Violin Plot with KDE and Box Plot overlay.

Example

The "width" of the violin corresponds to the height of the bars in a histogram. Look at the Benign violin's long, thin "needle" reaching up toward 5.0. This tells us that while cases above 3.0 are *statistically* outliers, they follow a continuous (though sparse) distribution.

8.4 Which Plot Should You Use?

gray!20 **Plot Type**	**Best Used For...**	**Main Weakness**
Histogram	Identifying exact peaks and gaps.	Can look "messy" with too many groups.
Box Plot	Comparing medians and identifying outliers.	Hides the "shape" (e.g., bimodal data).
Violin Plot	Detailed comparison of distribution shapes.	Can be harder for non-technical audiences.

Active Recall – Final Synthesis

If you were presenting to a doctor who only cares about the "typical" error value, which plot would you choose? If you were a researcher looking for hidden sub-patterns in the data, which would be better?

8.5 Practical Comparison: Which Spread to Trust?

Example

Look back at your **Texture Error** data for Benign cases (Pink).

- The **Standard Deviation** will be inflated by the outliers near 5.0.
- The **IQR** (the height of the pink box in your plot) remains $1.492 - 0.795 = 0.697$. This value stays the same even if that highest outlier moved from 5.0 to 100.0.

Note

Rule of Thumb: If your data looks like a "Bell Curve," use Standard Deviation. If your data has long tails or many outliers (like housing prices or medical errors), use IQR or Median AD.

9 Exploring Binary and Categorical Data

In the previous module, we focused on numeric variables like *texture error*. However, much of data science involves **Categorical Data**—discrete labels like "Diagnosis," "City," or "Property Type."

9.1 Proportions and Percents

The primary way we summarize categorical data is through **Proportions**. If our Breast Cancer dataset has 569 records and 212 are Malignant, the proportion is:

$$P(\text{Malignant}) = \frac{212}{569} \approx 0.37$$

Definition

Mode: The category that appears most often. Unlike the mean or median, the mode is the only estimate of location applicable to categorical data.

9.2 The Expected Value

In many data science applications, categories are tied to monetary values (e.g., the cost of a medical test). The **Expected Value** is a weighted average:

$$EV = \sum_{i=1}^{n} P(x_i) \times V_i$$

where $P(x_i)$ is the probability and V_i is the value.

Active Recall – Practical Logic

If a diagnostic test for Benign tumors costs $100 and a test for Malignant tumors costs $500, how would you use the proportions calculated above to find the "Expected Cost" per patient in your dataset?

10 Covariance and Correlation

In data science, variables rarely exist in isolation. To build predictive models, we must move from describing single features to understanding how variables move together through **Correlation**.

10.1 Covariance: The Direction of Change

The foundation of relationship measurement is **Covariance**, which indicates the joint variability of two random variables.

- **Positive Covariance**: Both variables tend to increase or decrease at the same time.
- **Zero Covariance**: The variables are independent of one another.
- **Negative Covariance**: As one variable increases, the other tends to decrease.

The Scaling Problem

Covariance can range from $-\infty$ to $+\infty$. Because the value is dependent on the units of measurement (e.g., meters vs. centimeters), it is extremely difficult to put these numbers into perspective or compare the "strength" of relationships across different datasets.

10.1.1 Covariance Formulas

Definition

Sample Covariance Formula:

$$S_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Population Covariance Formula:

$$\sigma_{xy} = \frac{\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)}{N}$$

—

10.2 Correlation: The Standardized Solution

Correlation is introduced specifically to solve the scaling problem of covariance. Unlike covariance, correlation is a **standardized measure** that takes on values strictly between -1 **and** $+1$. This makes it easy to interpret results regardless of the original data's units.

10.2.1 General Correlation Formulas

To interpret the relationship strength regardless of scale, we use the following standardizations:

Definition

Sample Correlation Formula (r):

$$r = \frac{S_{xy}}{S_x S_y}$$

Population Correlation Formula (ρ):

$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

- **A correlation of 1**: Known as perfect positive correlation; one variable perfectly explains the other.
- **A correlation of 0**: Means the variables are independent.
- **A correlation of -1**: Known as perfect negative correlation; they move perfectly in opposite directions.

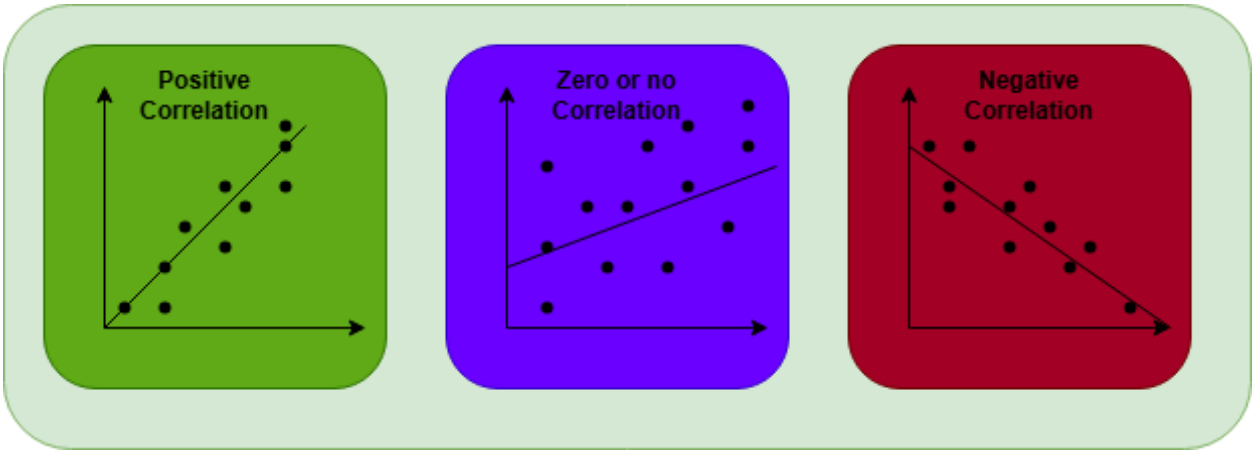

Figure 9: Visualizing the three primary types of correlation: Positive, Zero, and Negative.

—

10.3 Types of Correlation Coefficients

Correlation coefficients come in various forms, each suited to different types of data and relationships. These coefficients vary based on the type of relationship, measurement levels, and data distribution.

- **Pearson's** r: The most popular type, widely used for measuring **linear relationships** between two quantitative variables.
- **Spearman's** ρ: A non-parametric alternative for **ordinal** or non-normally distributed data; it analyzes rank-ordered variables.
- **Kendall's** τ: Another non-parametric option, often favored for smaller sample sizes due to its robustness.
- **Point-Biserial**: Used when one variable is dichotomous (binary) and the other is quantitative.
- **Cramér's V**: Applicable for measuring the correlation between two nominal variables.

—

10.4 Pearson's Correlation Coefficient (r)

Pearson's r describes the linear relationship between two continuous variables. It measures strength and direction, ranging from -1 to 1. It requires assumptions such as independence, interval/ratio scales, linearity, and normal distribution.

10.4.1 Manual Calculation Formula

For n data pairs, Pearson's r is calculated as:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

—

10.4.2 Numerical Example: Age vs. Income

To understand how Pearson's r is calculated, consider the relationship between Age (x) and Income (y). By organizing the data into a table, we can easily calculate the required sums for the formula.

Person	Age(x)	Income(y)	xy	x2	y2
1	20	1500	30000	400	2250000
2	25	2500	625000	625	6250000
3	30	3000	90000	900	9000000
4	40	5000	200000	1600	25000000
5	50	7500	3750000	25000	56250000
Total	165	19500	757500	6025	99000000

n=5 r=0.988
Strong positive correlation

Figure 10: Step-by-step calculation for Pearson Correlation between Age and Income.

Example

Using the totals from the table above where $n = 5$:

- $\sum x = 165$
- $\sum y = 19,500$
- $\sum xy = 757,500$
- $\sum x^2 = 6,025$
- $\sum y^2 = 99,000,000$

Substituting these into the formula results in $r \approx 0.988$. This value is very close to $+1$, indicating a **strong positive linear relationship**. As age increases, income increases in a nearly perfect linear fashion.

10.5 Spearman's Rank Correlation Coefficient (ρ)

Spearman's ρ is a non-parametric alternative used when data does not meet Pearson's assumptions. It ranks data points and measures how well the relationship can be modeled by a **monotonic function**—where variables change together but not necessarily at a constant rate.

This difference is clearly illustrated by the distinct behavior of the graphs below:

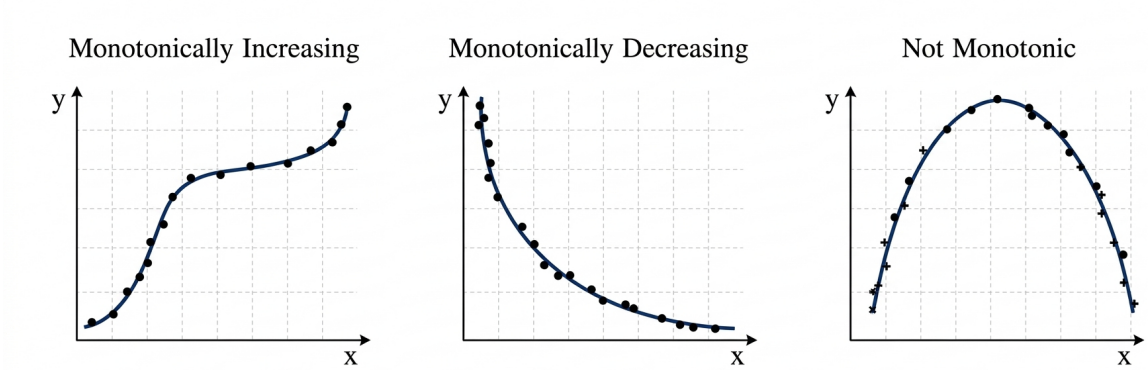

Figure 11: Illustrations of monotonic relationships versus a non-monotonic relationship.

10.5.1 Spearman Formula

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where d_i is the difference between the ranks of each pair of observations ($d_i = R(x_i) - R(y_i)$).

—

10.5.2 Numerical Example: History vs. Geography

To demonstrate, we will calculate the Spearman correlation between scores for 9 students in History and Geography.

Step 1: Rank the Data and Calculate Differences

First, the raw scores are ranked from highest to lowest, where a rank of "1" is assigned to the highest score. If scores are tied, they receive the average of the ranks they would otherwise occupy.

We then calculate the difference (d) between the ranks for each pair of observations using the following formula:

$$d_i = R(x_i) - R(y_i) \tag{1}$$

Where:

- $R(x_i)$ is the rank of the i-th observation of the first variable.
- $R(y_i)$ is the rank of the i-th observation of the second variable.

Once the differences are found, we calculate the squared difference (d^2) for each student to ensure all values are positive before summation.

History	Rank	Geography	Rank	d	d2
35	3	30	5	2	4
23	5	33	3	2	4
47	1	45	2	1	1
17	6	23	6	0	0
10	7	8	8	1	1
43	2	49	1	1	1
9	8	12	7	1	1
6	9	4	9	0	0
28	4	31	4	0	0

n=2 **rs=0.9**

Figure 12: Step-by-step ranking, difference (d), and squared difference (d^2) calculation for History and Geography scores.

We now have the essential components for our formula:

- Sum of d^2: $\sum d_i^2 = 4 + 4 + 1 + 0 + 1 + 1 + 1 + 0 + 0 = 12$
- Number of Data Pairs (n): 9

Step 2: Calculate the Coefficient

Substitute the calculated values ($\sum d_i^2 = 12$ and $n = 9$) into the Spearman formula:

$$
\begin{aligned}
r_s &= 1 - \frac{6\sum d_i^2}{n(n^2-1)} \\
r_s &= 1 - \frac{6(12)}{9(9^2-1)} \\
r_s &= 1 - \frac{72}{9(81-1)} \\
r_s &= 1 - \frac{72}{720} \\
r_s &= 1 - 0.1 \\
\mathbf{r_s} &= \mathbf{0.9}
\end{aligned}
$$

Interpretation: A coefficient of **0.9** indicates a **strong positive correlation** between History and Geography scores. This suggests that students who tend to score well in History also tend to perform well in Geography, even if the relationship is monotonic rather than strictly linear.

11 Advanced Correlation Analysis

In complex datasets like the Breast Cancer set, relying on a single metric can be misleading. A robust analysis requires moving from granular data visualization (Scatter Plots) to high-level summaries (Heatmaps).

11.1 Verification via Scatter Plots

Before generating a correlation matrix, it is essential to visualize the data distribution. While a heatmap provides a single numerical coefficient, a scatter plot reveals the "truth" behind that number.

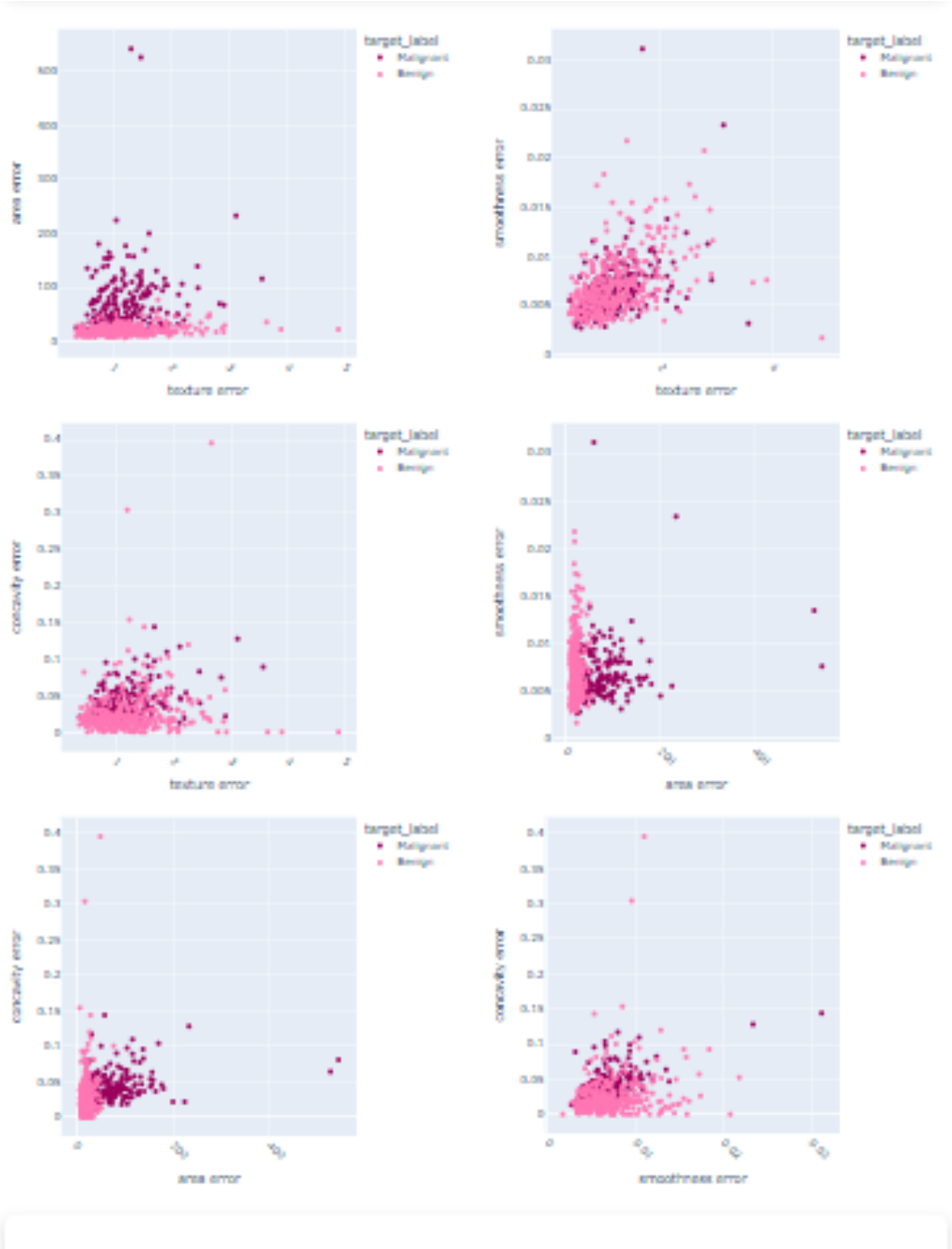

Figure 13: Scatter plot matrix for Breast Cancer features. These plots help identify outliers and verify if a relationship is linear or monotonic.

Data Science Insights

- **Redundancy**: If two features have a correlation near 1.0, they are redundant; we often remove one to simplify models.
- **Verification**: Always verify a high correlation with a **Scatterplot** to ensure the relationship isn't caused by an extreme outlier.

Why Scatter Plots First?

- **Outlier Detection**: Extreme values can artificially inflate Pearson's r, leading to false conclusions about feature importance.
- **Class Separation**: Visualizing "Malignant" vs. "Benign" labels helps identify which feature pairs are best for classification.
- **Relationship Shape**: It distinguishes between a straight-line (linear) and a curved (monotonic) relationship.

11.2 The Correlation Heatmap

Once the relationships are verified, a heatmap is used to identify **Redundancy**. If two features have a correlation near 1.0, they are redundant, and one may be removed to simplify the model.

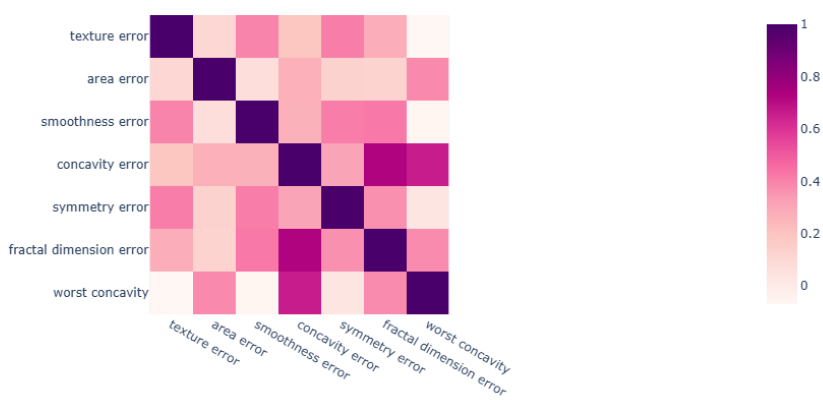

Figure 14: Seaborn Correlation Heatmap showing the global relationship structure of the dataset.

11.3 Choosing the Best Method: Pearson vs. Spearman

When calling `sns.heatmap(data.corr(method='...'))` in Python, students must choose the coefficient based on the patterns observed in the previous scatter plots:

Feature	**Pearson Correlation**	**Spearman Correlation**
Best Used For	Linear relationships.	Monotonic relationships.
Distribution	Normal (Bell Curve).	Any distribution (Non-parametric).
Outliers	Highly sensitive; can be distorted.	Robust; less affected by extreme values.
Breast Cancer Example	Useful for "Radius" vs "Perimeter" (highly linear).	Better for "Area Error" vs "Concavity" (often skewed).

Table 4: Comparison guide for selecting correlation methods in Seaborn heatmaps.

Data Science Insight

For the Breast Cancer dataset, many features are highly skewed. Using the **Spearman** method for your heatmap often provides a more reliable overview of feature interactions because it is less sensitive to the outliers visible in the scatter plots.

11.4 Correlation vs. Causation

A high correlation coefficient indicates a relationship between two variables, but it **does not prove** that one variable causes the change in the other.

The Golden Rule of Data Analysis

Correlation $\neq$ Causation. Two variables can be highly correlated due to:

- **Coincidence**: The variables happen to move together by pure chance.
- **Confounding (Lurking) Variables**: A third, unobserved variable is actually influencing both. For example, ice cream sales and shark attacks are correlated, but both are caused by a third variable: *Summer Heat*.

12 Exploring Two or More Variables

While correlation coefficients like Pearson's r and Spearman's ρ provide a numerical summary of relationships, visual displays are essential for uncovering the underlying structure of the data.

12.1 Visualizing Numeric vs. Numeric Data

Standard scatterplots are effective for small datasets, but for larger datasets (thousands of records), points often overlap into a dense "cloud". To address this, we use density-based visualizations.

12.1.1 Hexagonal Binning

In hexagonal binning, records are grouped into hexagonal cells. The color of each hexagon indicates the number of records it contains, revealing the true density of the relationship.

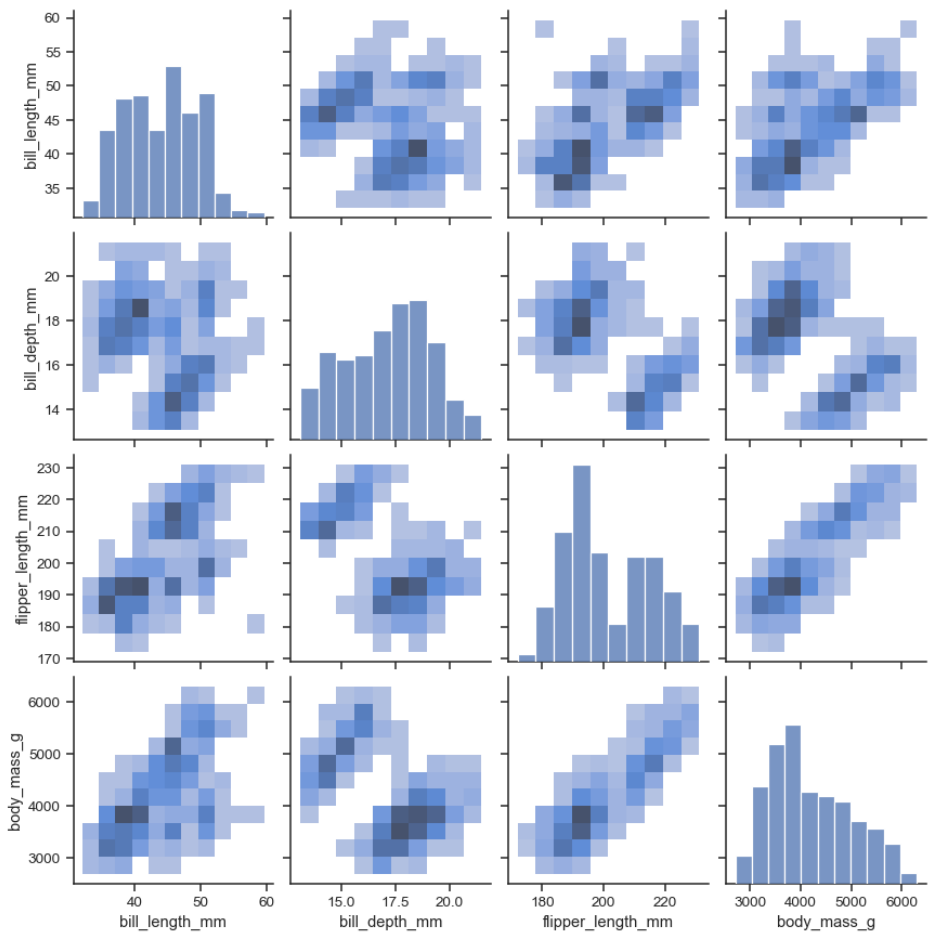

Figure 15: Hexagonal binning (represented here as rectangular density) clarifies the relationship between variables without the clutter of individual points [**?**].

12.1.2 Contour Plots (Topographical Maps)

Contour plots overlay a topographical map onto the data. Each contour band represents a specific density of points, peaking where data is most concentrated [**?**].

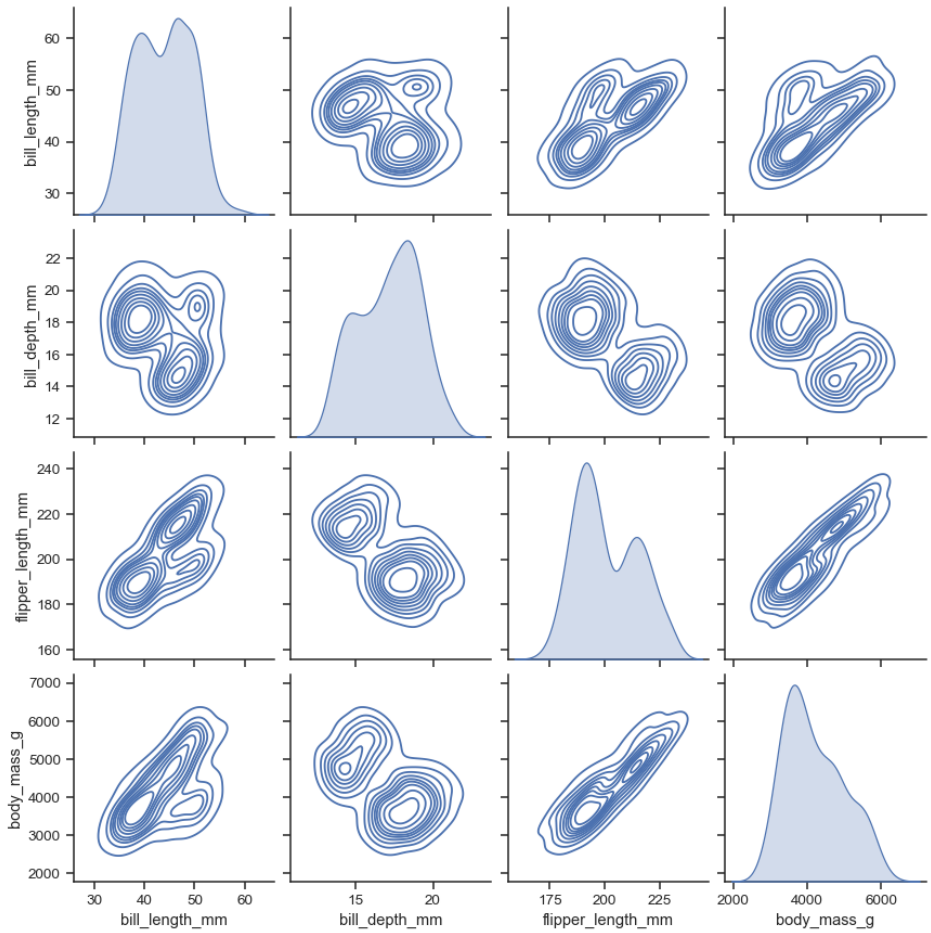

Figure 16: Contour plot showing 2D density peaks, a natural analog to the 1D density plot.

13 Multicollinearity and VIF

In predictive modeling, we often encounter **multicollinearity**—a situation where two or more predictor variables are highly correlated. This can make it difficult to determine the individual effect of each variable on the outcome.

13.1 Variance Inflation Factor (VIF)

We use the **Variance Inflation Factor (VIF)** to quantify the severity of multicollinearity. For a specific variable X_i, VIF is calculated using the coefficient of determination (R^2) from a regression of X_i against all other predictors:

$$VIF_i = \frac{1}{1 - R_i^2} \tag{2}$$

13.2 Interpretation Guide

- **VIF** ≈ 1: Low correlation; the variable is independent.
- **VIF** > 5: Moderate multicollinearity; requires investigation.
- **VIF** > 10: High multicollinearity; the variable is likely redundant and should be removed to improve model stability.

Data Science Tip

High VIF doesn't always mean a variable is "bad," but it does mean your model's coefficients are unstable. Removing high-VIF variables often leads to a simpler, more interpretable model.

14 Data and Sampling Distributions

Introduction to Sampling in the Era of Big Data

A popular misconception holds that the era of Big Data means the end of a need for sampling. In fact, the proliferation of data of varying quality reinforces the need for sampling as a tool to work efficiently and minimize bias. Even in Big Data projects, predictive models are typically developed and piloted with samples.

14.1 Population versus Sample

In statistics, the **population** is assumed to follow an underlying but unknown distribution. The only thing available to the researcher is the **sample data** and its empirical distribution. Modern statistics has moved toward focusing on the sample side where strong theoretical assumptions about the population are not required.

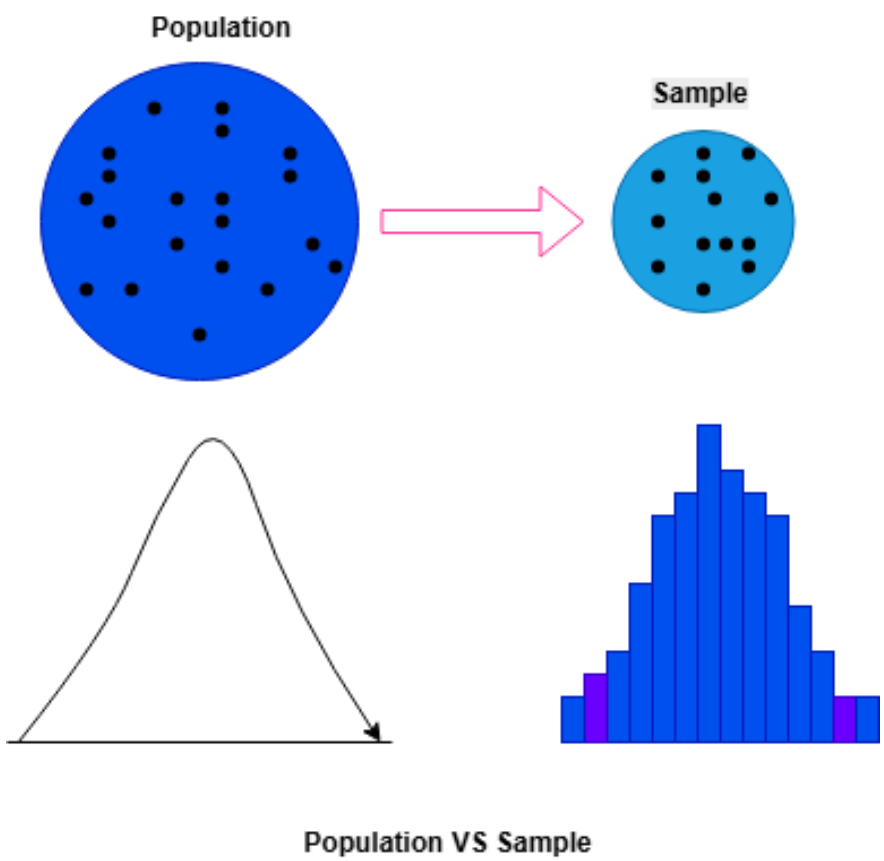

Figure 17: Schematic of Population (theoretical) vs. Sample (observed). The population follows an underlying unknown distribution, while the sample provides an empirical distribution.

15 Random Sampling and Sample Bias

A **sample** is a subset of data from a larger dataset, which statisticians call the **population**. **Random sampling** is a process where each member of the population has an equal chance of being chosen.

15.1 Key Terms

- **Sample**: A subset from a larger dataset.
- **Population**: The larger dataset, or the theoretical idea of a dataset.
- **Stratified Sampling**: Dividing the population into strata and randomly sampling from each.
- **Sample Bias**: A sample that misrepresents the population in a meaningful, non-random way.

15.2 The Importance of Data Quality

Data quality often matters more than data quantity when making an estimate. The 1936 *Literary Digest* poll failed despite 10 million respondents because it suffered from **sample bias**, polling only those with high socio-economic status. In contrast, George Gallup used a random sample of just 2,000 to accurately predict the election outcome.

16 Selection Bias and Regression to the Mean

16.1 Selection Bias

Selection bias occurs when data is selectively chosen—consciously or unconsciously—leading to misleading or ephemeral conclusions. This is often a result of **data snooping** or the **massive search effect**, where extensive hunting through data eventually uncovers a chance outlier that is mistaken for a true phenomenon.

16.1.1 Bias vs. Random Error

An important distinction must be made between random chance and systematic bias. Consider the analogy of a gun shooting at a target:

- **Unbiased Process**: Produces random error that does not tend strongly in any single direction.

- **Biased Process**: Produces systematic error where results consistently fall in a specific direction (e.g., the upper right quadrant).

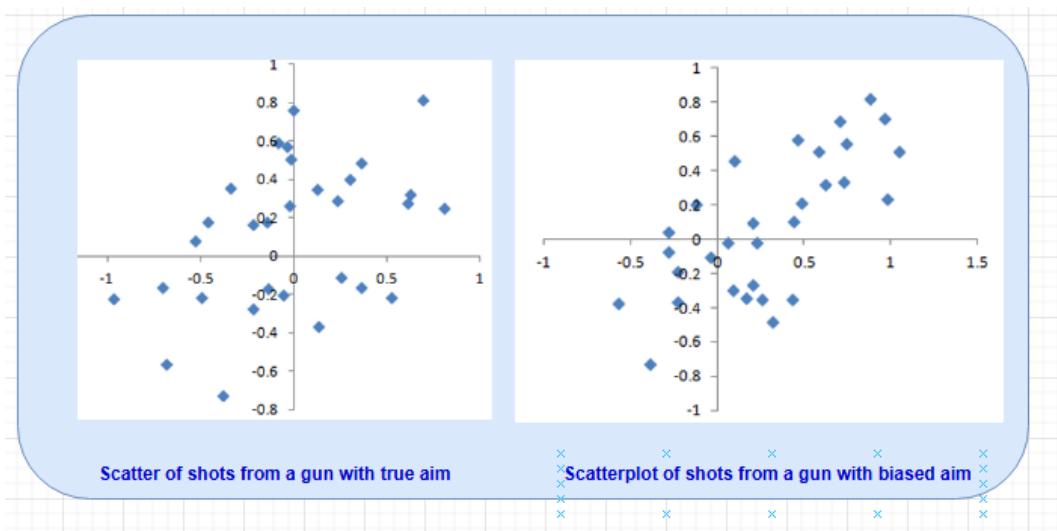

Figure 18: Unbiased scatter with true aim (left) vs. biased scatter with systematic error (right).

16.2 Regression to the Mean

Regression to the mean occurs when extreme observations are followed by more central ones. For example, a "Rookie of the Year" often experiences a "sophomore slump" because their initial extreme performance was a combination of skill and luck; while the skill remains, the luck usually regresses to the average.

17 Sampling Distribution of a Statistic

The **sampling distribution** refers to the frequency distribution of a sample statistic (like the mean) over many samples drawn from the same population.

17.1 The Central Limit Theorem (CLT)

The **Central Limit Theorem** is a fundamental pillar of statistics. it states that the distribution of sample means will be nearly normal, regardless of the population's shape, provided the sample size is large enough [**?**, **?**].

17.2 Visualizing Convergence

As shown in Figure 19, the "before and after" effect of the CLT is striking. Even when starting with heavily skewed or uniform populations, the sampling distribution becomes symmetric and bell-shaped as n moves from 10 to 30.

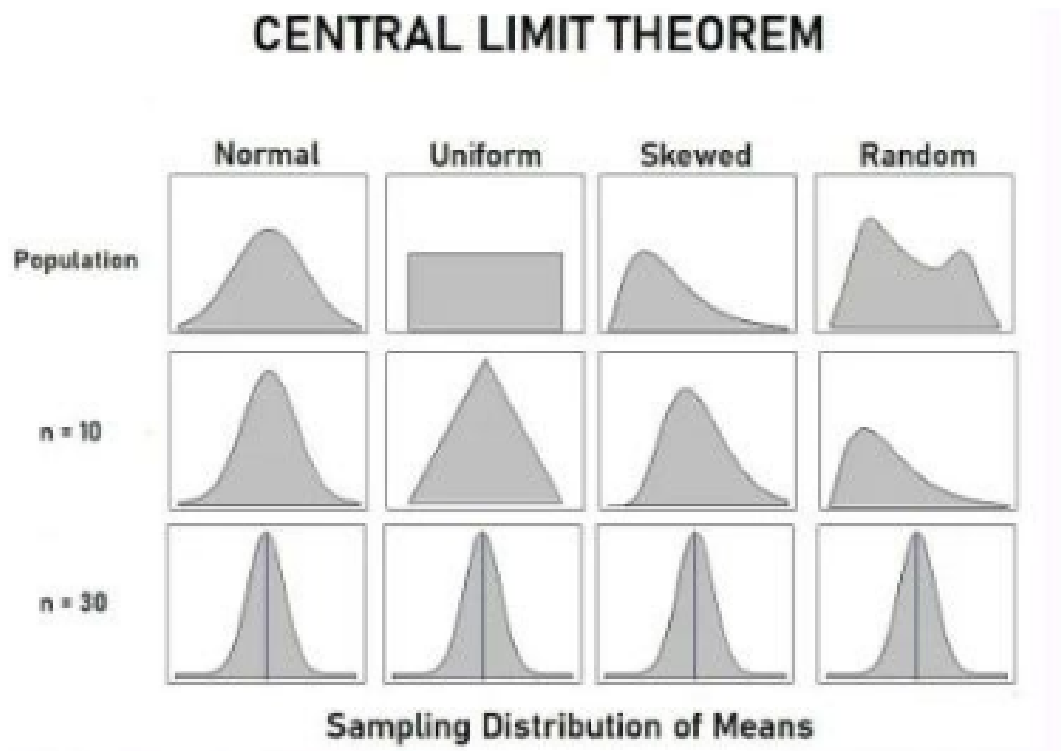

Figure 19: The Central Limit Theorem in action: Different population shapes all converge toward a Normal sampling distribution as n increases.

17.3 The Mathematical Formula

This visual convergence is governed by the following relationship, which defines the center and spread of the resulting normal distribution:

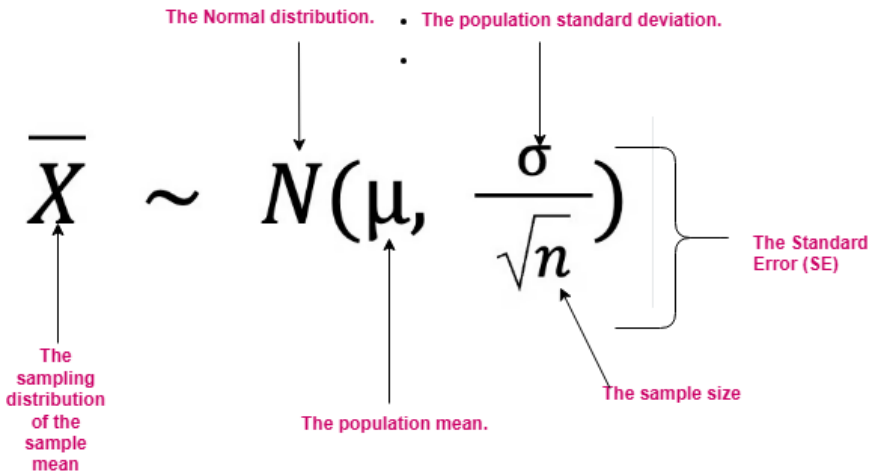

Figure 20: Component breakdown of the CLT equation.

Where:

- $\bar{X}$: The sampling distribution of the sample mean [**?**].
- N: The Normal distribution, which the sampling distribution follows as n increases [**?**].
- μ: The population mean (the theoretical center of the distribution) [**?**].
- σ: The population standard deviation (the spread of the original population).
- n: The sample size (the number of observations in each sample) [**?**].
- $\frac{\sigma}{\sqrt{n}}$: The **Standard Error (SE)**, which represents the variability of the sample statistic over many samples [**?**].

Key Ideas

1. Random sampling remains essential even in Big Data to ensure representativeness.
2. Data quality is often more important than quantity.
3. Distinguish between *data distribution* (individual points) and *sampling distribution* (sample statistics).

18 The Bootstrap

The bootstrap is a powerful, computationally intensive tool used to assess the variability of a sample statistic without making strong assumptions about the underlying distribution.

18.1 Concept and Procedure

Conceptually, the bootstrap involves treating the original sample as a "hypothetical population" and drawing new samples from it with replacement.

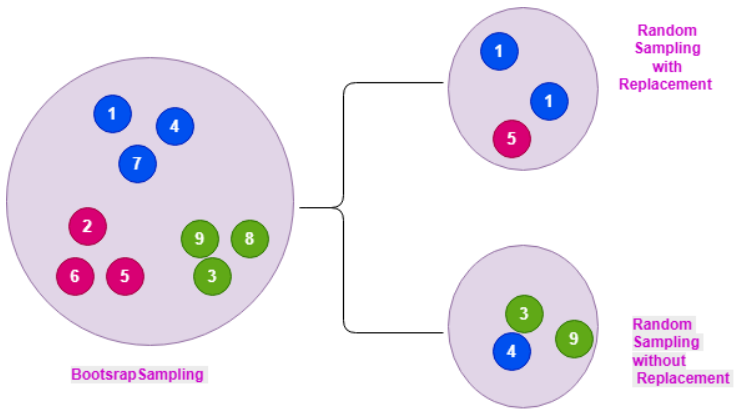

Figure 21: The Bootstrap idea: Generating a sampling distribution by repeatedly resampling the observed data

Important Note

The bootstrap does **not** create new data or compensate for a small sample size. It merely informs us about how additional samples would behave if they were drawn from a population identical to our original sample [**?**]

18.2 Bootstrap Aggregating (Bagging)

In data science, the bootstrap is used beyond simple statistics. For example, in predictive modeling, multiple models can be run on different bootstrap samples. Averaging their predictions is a process called **bagging** (bootstrap aggregating), which generally outperforms a single model.

19 Confidence Intervals

While point estimates (single numbers) are common, they fail to account for uncertainty. **Confidence intervals** present estimates as a range, grounded in statistical sampling principles to counteract the human tendency to place undue faith in a single number.

19.1 Visualizing the Interval

As shown in Figure 22, a confidence interval (such as 95%) encloses the central portion of the sampling distribution. The higher the level of confidence required, or the smaller the sample size available, the wider this interval must become.

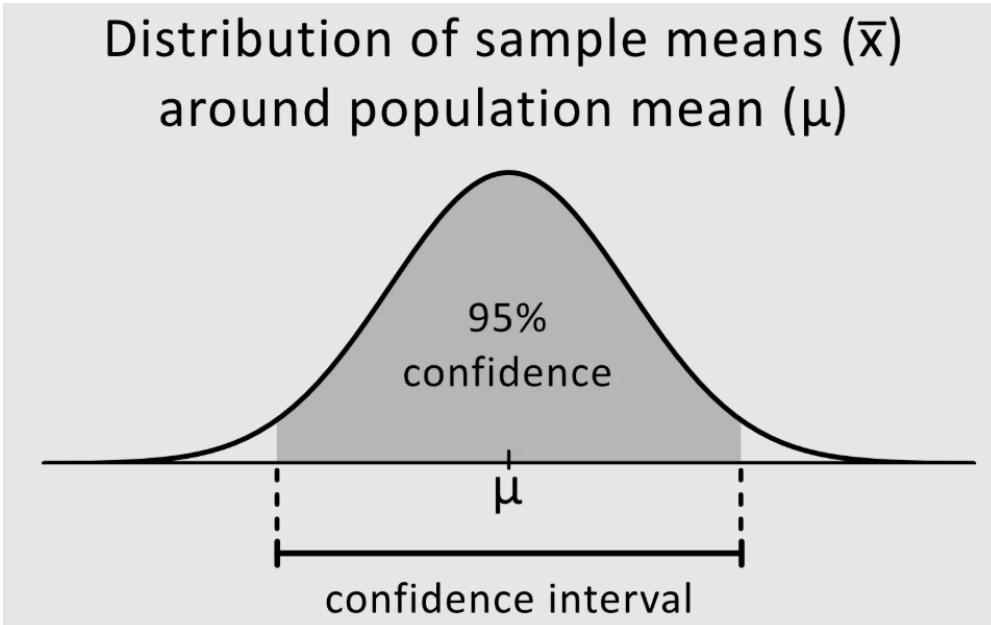

Figure 22: The distribution of sample means ($\bar{x}$) around the population mean (μ), illustrating a 95% confidence interval range.

19.2 Key Drivers of Interval Width

The width of a confidence interval is influenced by three main factors:

- **Confidence Level**: Higher levels (e.g., 95% vs. 90%) result in wider intervals to be more assured of capturing the true value.
- **Sample Size (n)**: Smaller samples lead to wider intervals due to increased uncertainty.
- **Data Variability**: The more variable the underlying data, the wider the resulting interval.

Data Science Perspective

For data scientists, confidence intervals are primarily used to communicate the potential error in an estimate and to determine if more data (a larger sample) is required for a reliable result.

19.3 Key Ideas

- **Interval Presentation**: Confidence intervals are the typical way to present estimates as a range rather than a single point.
- **Bootstrap Application**: The bootstrap is a versatile, formula-free way to construct these intervals for almost any statistic.

20 Important Distributions in Data Science

Understanding the characteristic shapes and utilities of different distributions allows data scientists to better model real-world phenomena and recognize the underlying processes generating their data.

20.1 Key Distributions and Their Utility

Distribution	Visual Shape	Data Science Utility
Normal (Gaussian)	Symmetric, bell-shaped curve.	Used for the Central Limit Theorem, most sample means, and many machine learning algorithms (e.g., Linear Regression).
Log-Normal	Heavily right-skewed; starts at zero.	Ideal for modeling wealth, income, or any data where values cannot be negative and have long tails.
Binomial	Discrete bars representing successes/failures.	Used for binary outcomes, like "click vs. no-click" in A/B testing or "churn vs. retain."
Poisson	Discrete; skewed but becomes more symmetric as the rate increases.	Used for "arrival" data, such as the number of users visiting a website per hour or calls to a center.
Uniform	Flat; all values have the same probability.	Often used as a starting point for simulations or when every outcome is equally likely (like rolling a fair die).

Table 5: Common probability distributions and their applications in Data Science.

20.2 Important Distributions in Data Science

Identifying the characteristic shape of your data allows for better model selection and more accurate statistical inferences.

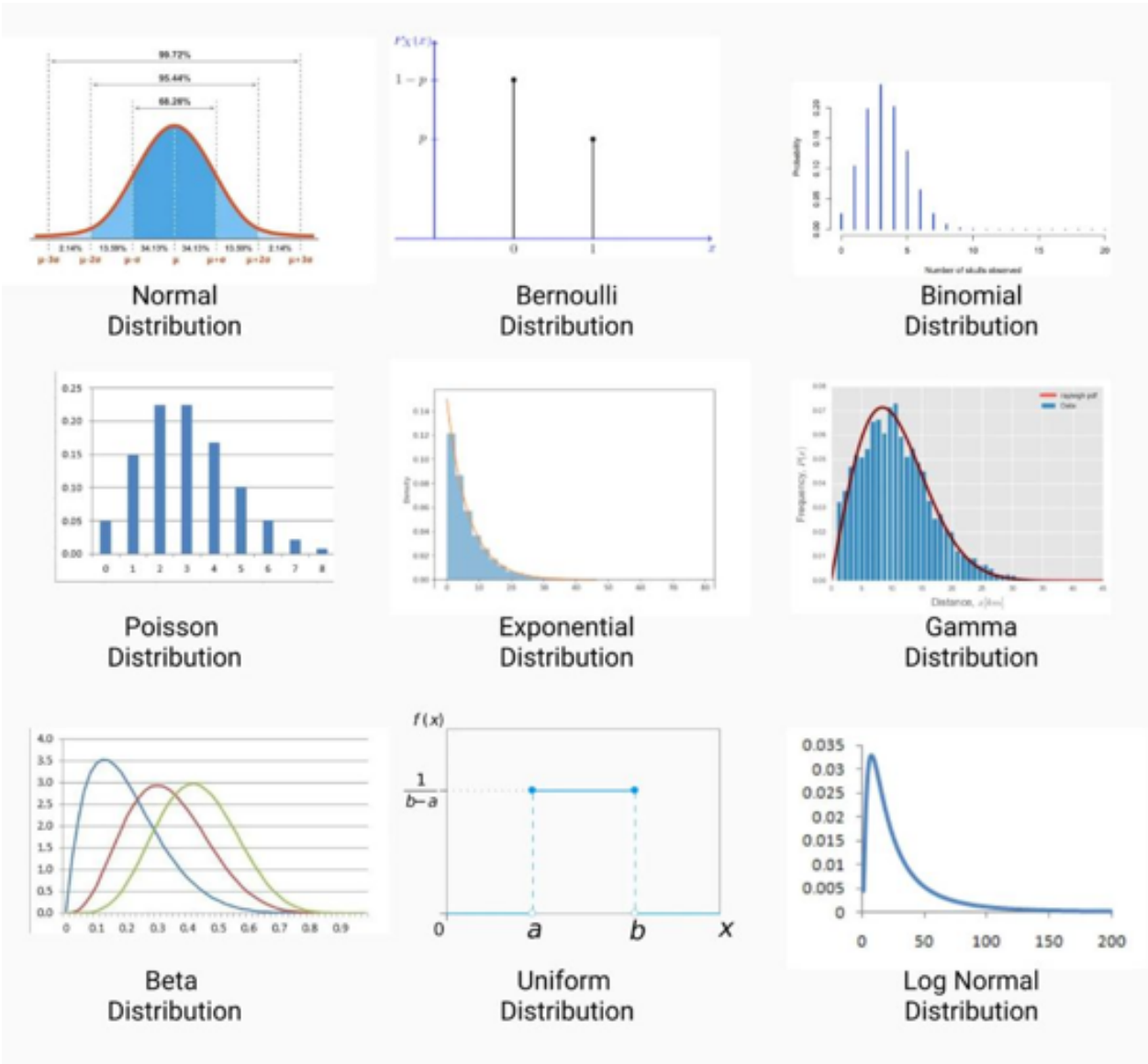

Figure 23: Visual comparison of common data distributions, including Normal, Binomial, Poisson, and Log Normal.

20.3 Key Ideas

- **Recognition**: Identifying a distribution's shape helps in choosing the correct statistical test or model.

- **Transformation**: If data is highly skewed (like Log-Normal), data scientists often apply a "Log Transformation" to make it look more Normal for better model performance.

20.4 Quantifying Difference: KL Divergence

Beyond identifying shapes, we often need to measure the mathematical "distance" between two distributions. **Kullback-Leibler (KL) Divergence** quantifies the information loss when one distribution (Q) is used to model another (P).

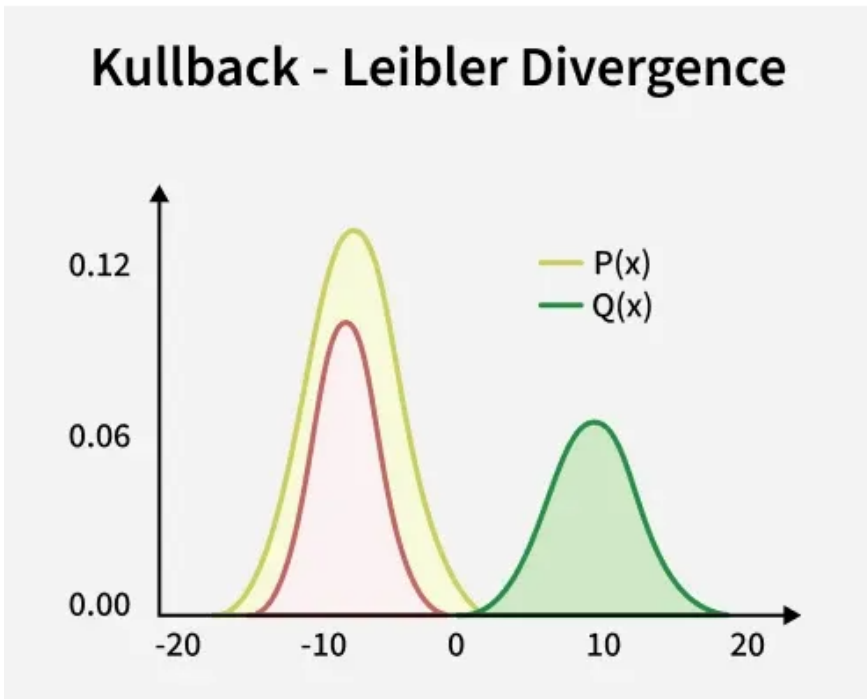

Figure 24: KL Divergence between distributions $P(x)$ and $Q(x)$.

- **Relative Entropy**: It measures the efficiency of using Q to represent P.
- **Non-Symmetric**: The divergence from P to Q is distinct from Q to P.
- **Applications**: Used in training neural networks and monitoring model decay over time.

21 Inferential Statistics: The Signal and the Noise

The goal of inferential statistics is to determine if a pattern in our sample is a **true signal** that exists in the whole population, or just **random noise**.

21.1 Why We Use Statistical Tests

Statistical tests act as a filter. They calculate the *p-value*, which is the probability that the noise alone could have created the pattern we see.

- **High p-value**: The pattern is likely just noise. We stay with the **Null Hypothesis** (H_0).
- **Low p-value** (≤ 0.05): The pattern is likely a real signal. We accept the **Alternative Hypothesis** (H_a).

21.2 Choosing Your "Signal Detector"

Goal	**Data Type**	**Recommended Test**
Compare 2 Groups	Numerical	**t-Test**
Compare 3+ Groups	Numerical	**ANOVA**
Find Relationships	Numerical	**Correlation / Regression**
Compare Buckets	Categorical	**Chi-Square** (χ^2)

Table 6: Quick reference for selecting the appropriate statistical test.

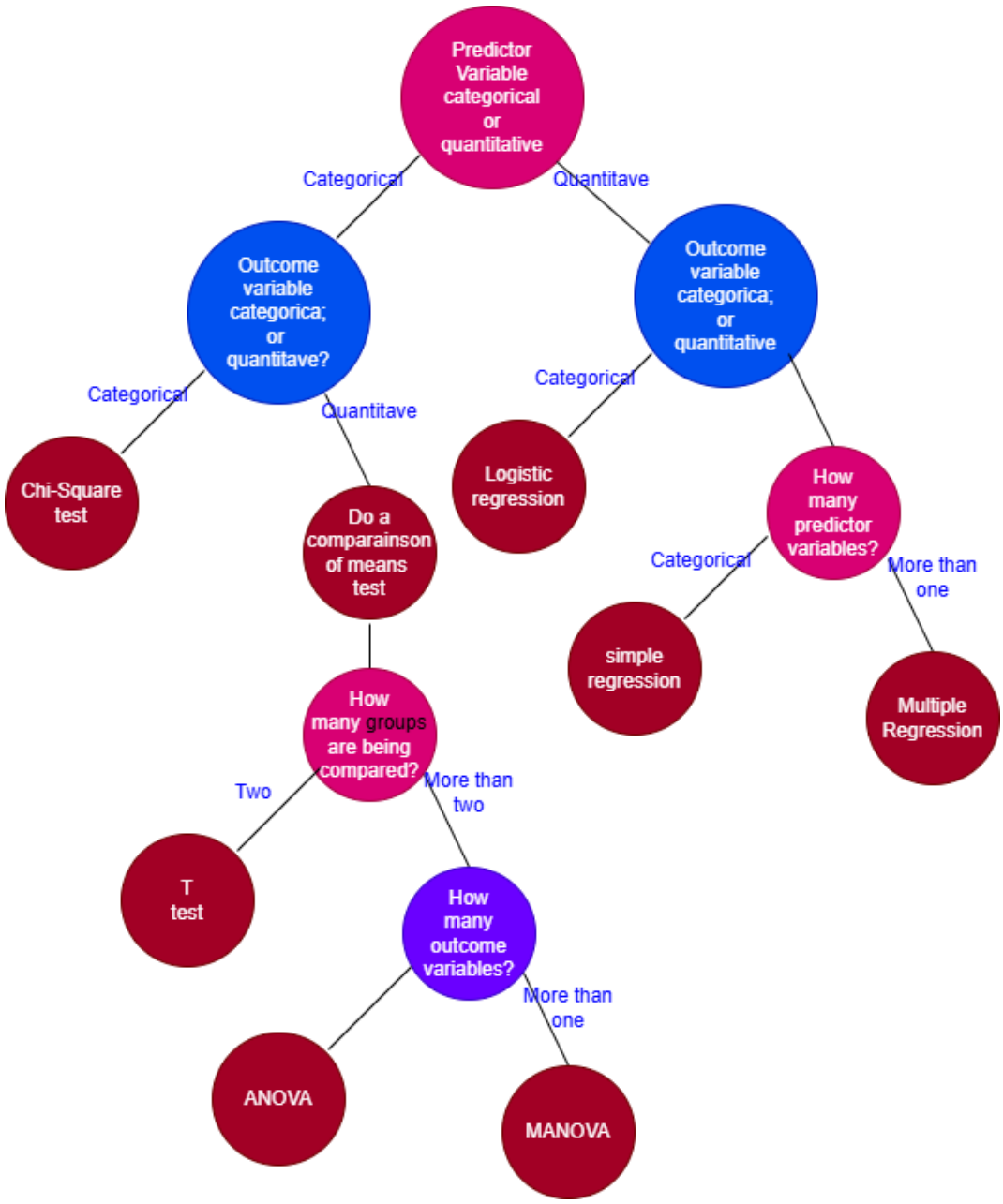

Figure 25: The full decision roadmap for inferential modeling.

22 Bayesian Statistics

Unlike Frequentist statistics, which relies on p-values and fixed parameters, **Bayesian Statistics** treats parameters as distributions and incorporates prior knowledge into the analysis.

22.1 Bayes' Theorem

The fundamental formula for updating beliefs based on new evidence is:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)} \tag{3}$$

- $P(H|E)$ **(Posterior)**: The probability of the hypothesis after seeing the evidence.
- $P(E|H)$ **(Likelihood)**: The probability of the evidence given the hypothesis.
- $P(H)$ **(Prior)**: Our initial belief before seeing the data.

22.2 Utility in Data Science

- **Small Data**: Bayesian methods are powerful when data is scarce because the "Prior" helps stabilize the estimates.
- **A/B Testing**: Allows for "continuous monitoring" of results without the "peeking" penalty associated with Frequentist p-values.
- **Spam Filtering**: Naive Bayes classifiers use these principles to predict if an email is spam based on word frequencies.